anti-dervatives or integrals

anti-dervative

$F(x) = \frac{1}{1+x^2}$ $\qquad \int \tan^{-1} x + C$

$F(x) = e^x$ $\qquad \int e^x + C$

$F(x) = a^x$ $\qquad \int \frac{a^x}{\ln(a)} + C$

$F(x) = \frac{1}{x}$ $\qquad \int \ln|x|$ because $\frac{1}{x} = x^{-1}$ which can't happen

$F(x) = x^N$ $\qquad \int \frac{x^{N+1}}{N+1}$ $\quad N \neq -1$

example:

$F(x) = 5x^4 - 6x^2 + 3x - 5$

$\int 5\frac{x^{4+1}}{4+1} - 6\frac{x^{2+1}}{2+1} + 3\frac{x^{1+1}}{1+1} - 5\frac{x^{0+1}}{0+1}$

a x^0 is just one or another way 1 can be written so if it is there you must take the anti-dervative

$f(x) = x^5 - 2x^3 + \frac{3}{2} - 5x + C$

the anti-dervative is the apposite of the dervative you can think of the anti-dervative as undoing what the dervative is doing

if you finish taking the antidervative and want to check if you did it correctly you can take the dervative you should get the orginal question example:

$$F(x) = \frac{6x^4 - 3x^2 + 2x - 5}{6x^4} \Rightarrow \frac{\cancel{6x^4}}{\cancel{6x^4}} - \frac{3x^2}{6x^4} + \frac{2x}{2x^4} - \frac{5}{6x^4}$$

$$F(x) = 1 - \frac{1}{2x^2} + \frac{1}{x^3} - \frac{5}{6x^4} \Rightarrow \int \frac{x^{0+1}}{0+1} + \frac{1}{2}\frac{x^{-2+1}}{-2+1} + \frac{x^{-3+1}}{-3+1} - \frac{5}{6}\frac{x^{-4+1}}{-4+1}$$

$$\boxed{F(x) = x - \frac{1}{2x} - \frac{x^2}{2} + \frac{5}{18x^3} + C}$$

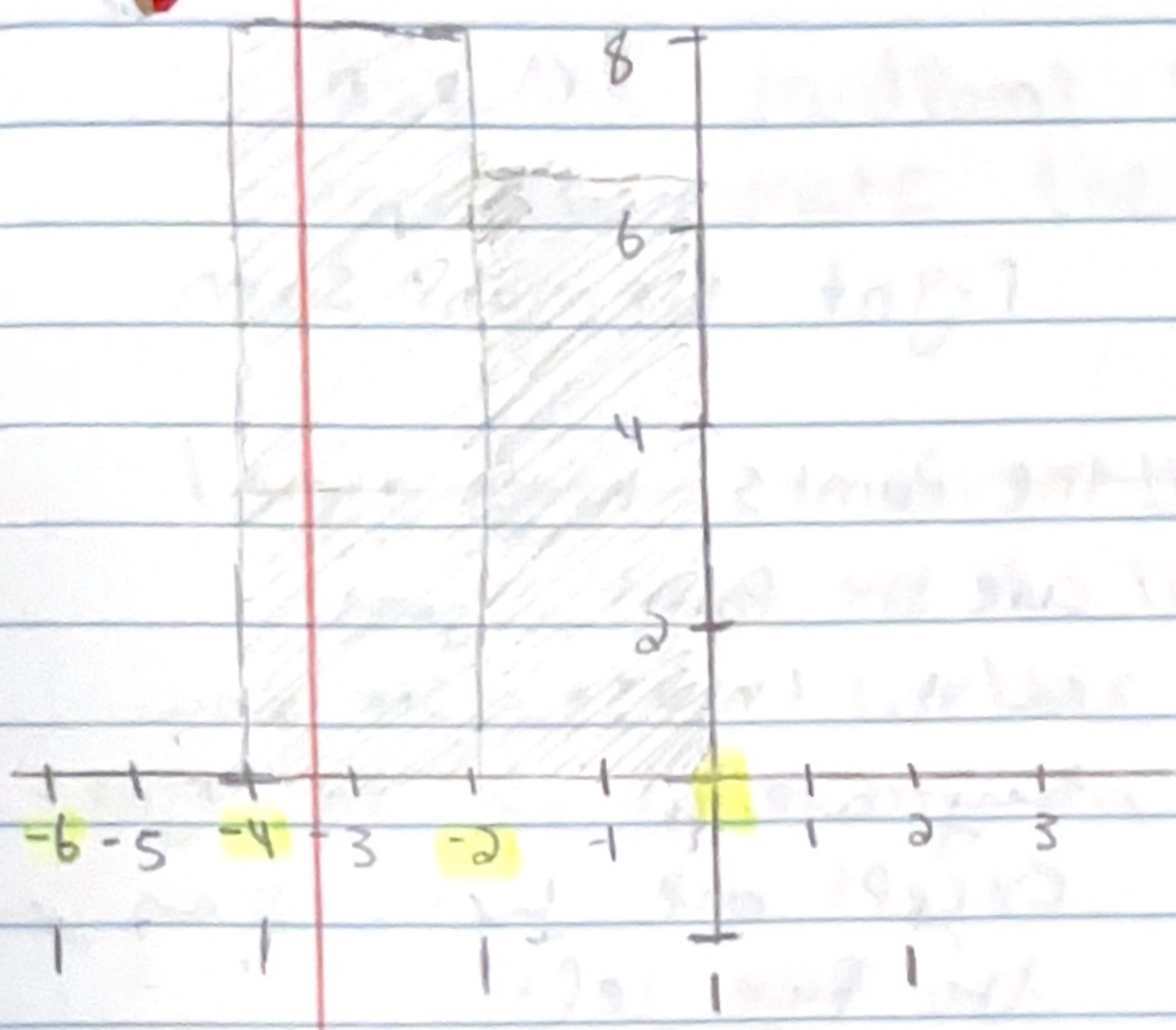
there is one term in the denominator so you can break it up

$$F(x) = \frac{7}{x^2+1} - 8e^x + 5^x$$

$$\int \frac{7}{x^2+1} - 8e^x + 5^x \Rightarrow 7\tan^{-1}x - 8e^x + \frac{5^x}{LN(5)}$$

$$\boxed{7\tan^{-1}x - 8e^x + \frac{5^x}{LN(2)} + C}$$

find the left riemann sum of $f(x)$ in the interval $[-6, 2]$, $N=4$



$$\frac{b-a}{N}$$

$[-6, 2]$
a b

$\Delta x = \frac{2-(-6)}{4} \quad \frac{8}{4} = 2$

a is the smaller value

b is the bigger value

$\Delta x = 2$

(left) $x_0 = -6$

(Left) $x_1 = -4$

(left) $x_2 = -2$

(left) $x_3 = 0$

$x_4 = 2$

there are usally many to the specific direction either left or right except one being the last on the oppisite side of the given direction for this one all on the left work as left except 2 because it is not on the left picking it would make the whole thing but they asked for the left on a given interval

$f(x) = 0.1x(x^2-36)$

$f(-6) = 0.1(-6)((-6)^2-36) = 0$

$f(-4) = 0.1(-4)((-4)^2-36) = 8$

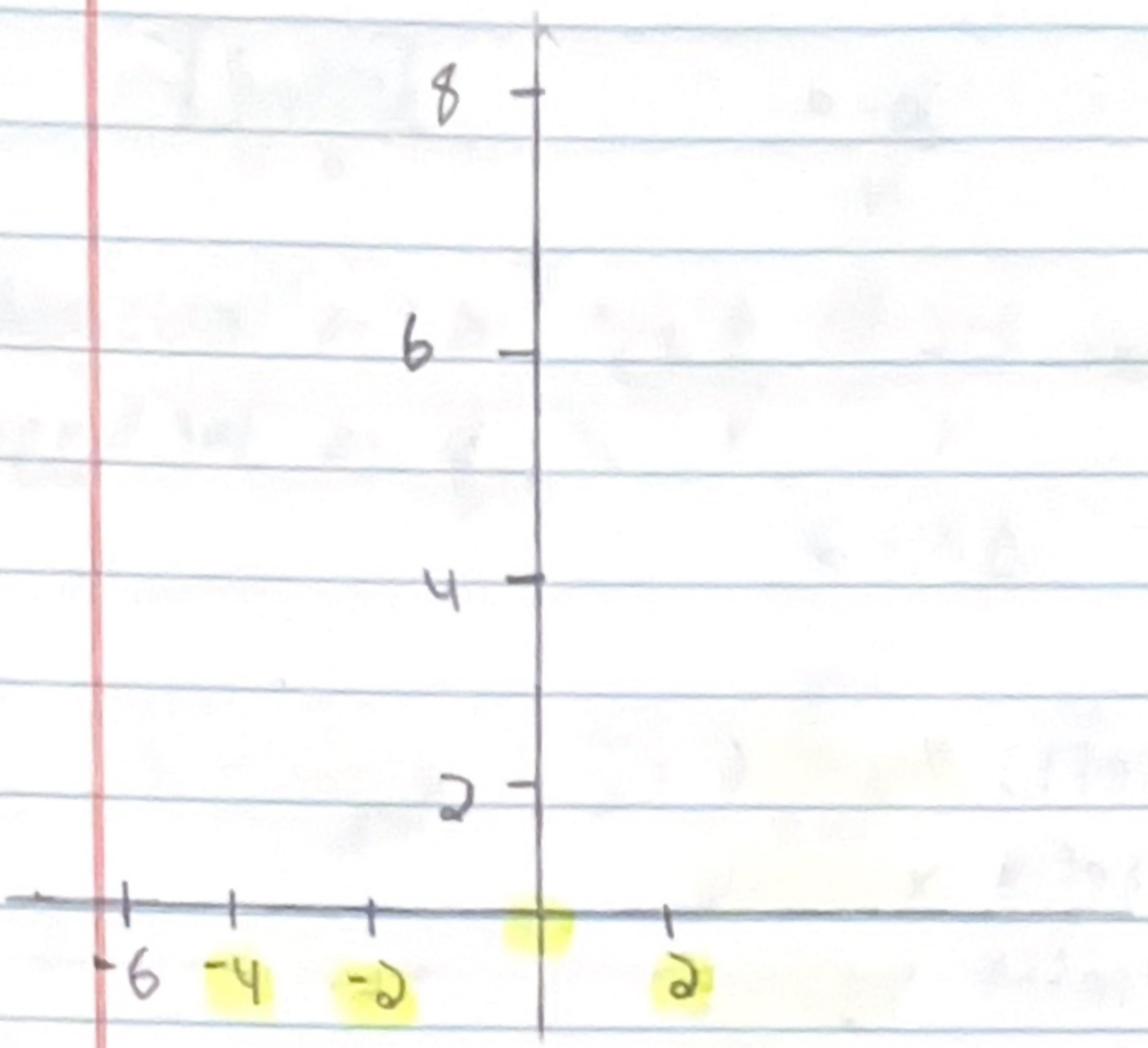
$f(-2) = 0.1(-2)((-2)^2-36) = 6.4$

$f(0) = 0.1(0)((0)^2-36) = 0$

$\Delta x[f(x_0) + f(x_1) + f(x_2) + f(x_3)]$

$2[-6 + -4 - 2 + 0] = \boxed{24}$

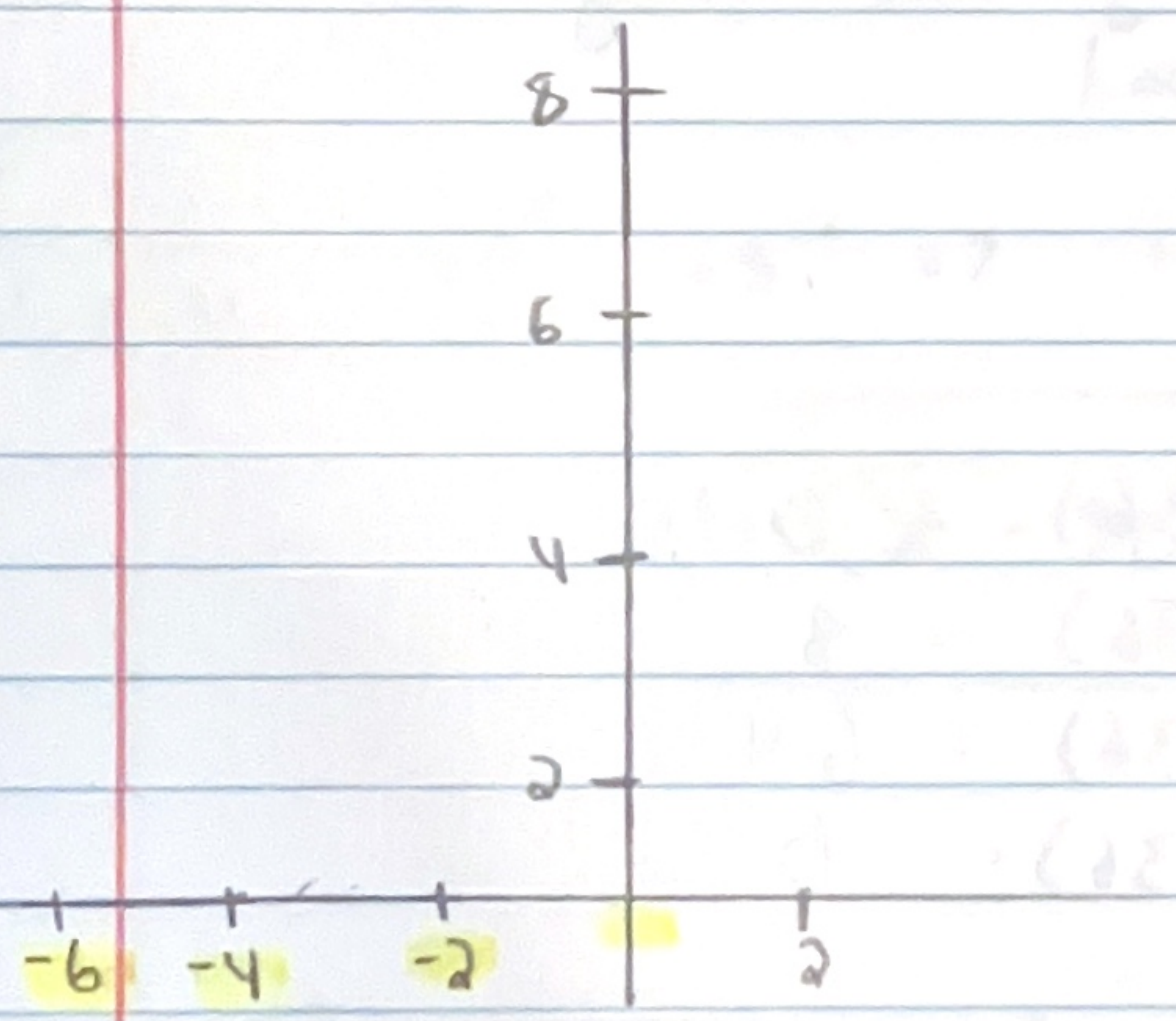
if it asked for the right riemann sum



right riemann sum

the points highlighted are the point you will choose see how they are all on the right except one with is on the far left

Same one from the last page



left riemann sum

the points that are highlighted are the point you will choose for the left riemann sum notice how all the points are on the left except one on the far right midPoint

use the midPoint riemann Sum with $N=5$
to approximate the integral $\int_0^1 \sqrt{x^2+1}$

$[0,1]$

a is the smaller value
b is the bigger value
wether on a interval or
on a integral

$$\Delta x = \frac{b-a}{N} \quad \frac{1-0}{5} = \frac{1}{5}$$

$x_0 = 0$

$x_1 = \frac{1}{5}$ — $\bar{x}_1$ $\quad (0+\frac{1}{5}) \div 2 = \frac{1}{10}$

$x_2 = \frac{2}{5}$ — $\bar{x}_2$ $\quad (\frac{1}{5}+\frac{2}{5}) \div 2 = \frac{3}{10}$

$x_3 = \frac{3}{5}$ — $\bar{x}_3$ $\quad (\frac{2}{5}+\frac{3}{5}) \div 2 = \frac{5}{10}$

$x_4 = \frac{4}{5}$ — $\bar{x}_4$ $\quad (\frac{3}{5}+\frac{4}{5}) \div 2 = \frac{7}{10}$

$x_5 = 1$ — $\bar{x}_5$ $\quad (\frac{4}{5}+1) \div 2 = \frac{9}{10}$

$$\Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4) + f(\bar{x}_5) \right]$$

$$\frac{1}{5}\left[\sqrt{1.01} + \sqrt{1.09} + \sqrt{1.25} + \sqrt{1.49} + \sqrt{1.81} \right]$$

$$= \boxed{1.1466}$$

Plug them back into
the given equation